```
D:\Software\Python\Python311\python.exe E:\PythonProjectsByYear\Year2025\MC-PIKAN_Official\experiments\Volterra1D_MCPI.py
The network model is cPIKAN
Layers: ModuleList(
  (0): ChebyKANLayer(in_features=1, out_features=10, degree=3)
  (1): ChebyKANLayer(in_features=10, out_features=10, degree=3)
  (2): ChebyKANLayer(in_features=10, out_features=1, degree=3)
)
Total trainable parameters: 480
Model on device: cuda:0
Start training...
Max Epochs: 2000, Learning Rate: 0.001, Epsilon: 1e-20, Optimizer: Adam
Training: 100%|██████████| 2000/2000 [00:08<00:00, 222.50epoch/s, Loss=5.15e-06, LR=1.0e-03, Best=5.15e-06]
Training finished. Best loss: 3.87e-06. Model saved to results/best_model.pth
Loss history saved to results/loss_history.npy
Training time: 9.757103204727173s
Relative loss: 0.002119426615536213
L2 loss: 0.001348594087176025
Parameter containing:
tensor([[[ 0.0755,  0.1375,  0.4770, -0.0733],
         [-0.0299, -0.5496, -0.2197,  0.1886],
         [-0.0745,  0.2990,  0.3424,  0.0309],
         [-0.0989, -0.2021, -0.0746,  0.7627],
         [ 0.1914, -0.1468, -0.5302, -0.0916],
         [-0.6144,  0.1440,  0.3437, -0.0600],
         [ 0.2836, -0.0094, -0.2529, -0.2300],
         [ 0.0134,  0.3409, -0.2979, -0.3087],
         [ 0.6774, -0.1723, -0.6363, -0.3819],
         [ 0.3059, -0.3293, -0.5617, -0.0768]]], device='cuda:0',
       requires_grad=True)
Parameter containing:
tensor([[[-7.1736e-03,  2.0965e-01,  1.4438e-02, -1.8432e-01],
         [ 5.0157e-02, -2.1889e-01, -6.7568e-02,  2.3730e-01],
         [ 2.5008e-02,  1.2324e-01, -2.0344e-02, -9.4660e-02],
         [ 1.2677e-02, -2.1773e-01, -2.8632e-02,  2.0083e-01],
         [-2.8669e-02,  2.0362e-01,  4.9188e-02, -1.4151e-01],
         [ 3.9111e-02, -1.8653e-01, -3.4125e-02,  1.8665e-01],
         [ 1.9120e-02, -1.4583e-01, -1.9238e-02,  1.3856e-01],
         [ 4.9963e-02, -2.1759e-01, -1.2563e-02,  1.5615e-01],
         [-2.4008e-02,  1.7547e-01, -1.5053e-02, -2.2512e-01],
         [-2.9764e-02,  2.3239e-01, -7.1166e-03, -1.6561e-01]],

        [[-4.3983e-02, -2.1387e-01,  5.1480e-02,  1.8583e-01],
         [ 3.4705e-02,  2.1490e-01, -8.2944e-03, -1.6680e-01],
         [ 3.9980e-02, -2.4171e-01, -1.1751e-02,  2.5556e-01],
         [ 4.8215e-02,  2.0823e-01, -4.9267e-02, -2.3362e-01],
         [-2.2888e-02, -2.1318e-01,  4.1693e-02,  1.8105e-01],
         [-1.6146e-02,  1.6664e-01, -6.1385e-02, -1.7319e-01],
         [ 4.5230e-02,  2.2284e-01,  1.5786e-03, -1.6061e-01],
         [-1.9640e-02,  2.1866e-01, -2.2558e-02, -2.0906e-01],
         [-6.1455e-03, -2.7291e-01, -4.0156e-03,  2.2060e-01],
         [-6.1345e-02, -2.0688e-01,  6.0601e-02,  1.8347e-01]],
```

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    [[-2.1496e-02,  1.5618e-01, -2.8892e-02, -1.7415e-01],
     [ 8.5741e-02, -2.8735e-01, -6.1892e-02,  2.7721e-01],
     [ 5.4633e-03,  9.2439e-02,  3.3840e-03, -3.6746e-02],
     [ 2.3029e-02, -2.2796e-01,  5.0629e-03,  2.0452e-01],
     [-8.4739e-02,  2.4465e-01,  6.1366e-02, -1.9771e-01],
     [ 5.1930e-02, -1.4375e-01, -3.1805e-02,  2.0173e-01],
     [ 2.4514e-02, -1.7960e-01, -1.6516e-02,  1.3424e-01],
     [ 3.4052e-02, -2.0581e-01, -3.6277e-02,  1.4737e-01],
     [-4.7161e-02,  2.0764e-01, -3.7228e-02, -2.0235e-01],
     [-7.5739e-03,  1.8510e-01,  1.2566e-02, -1.9953e-01]],

    [[-4.7562e-02, -6.8966e-02, -2.8935e-02, -8.6979e-03],
     [ 9.5818e-02, -5.9619e-02,  1.9663e-02, -8.0499e-02],
     [ 8.8447e-02, -1.5075e-01,  2.4073e-02,  1.1965e-01],
     [-2.3099e-03,  1.0243e-01,  1.0767e-02, -7.9396e-02],
     [-4.5183e-02, -8.0252e-04,  4.9461e-03,  2.4577e-02],
     [ 3.6341e-02, -1.3906e-02, -2.5349e-02, -2.2004e-02],
     [-3.4928e-03, -7.0063e-03, -3.5688e-02, -1.2788e-02],
     [ 2.4684e-02,  1.3549e-01, -5.9140e-02, -1.7310e-01],
     [-2.0211e-02, -6.5886e-03,  4.3911e-02,  1.2815e-02],
     [-1.3280e-02, -3.8747e-02,  5.9627e-02,  4.3820e-02]],

    [[-1.9592e-02, -6.7189e-02, -4.1217e-03,  3.9063e-02],
     [-4.7628e-03,  1.4053e-01, -2.1819e-02, -1.3295e-01],
     [ 7.3379e-03, -6.5909e-03, -3.4882e-02, -4.0775e-02],
     [ 8.2949e-03,  1.6429e-02, -9.7239e-03, -4.1096e-02],
     [-3.9013e-02, -6.0662e-02,  3.5831e-03,  8.0712e-02],
     [ 7.9104e-03,  1.1861e-01, -2.2597e-02, -2.9912e-02],
     [ 1.0502e-02,  3.0536e-02, -1.9442e-02, -8.5963e-02],
     [ 6.4282e-03,  8.2510e-02, -4.2313e-02, -6.2108e-02],
     [ 1.7754e-03, -7.6038e-02, -9.6204e-03, -7.3632e-03],
     [-2.8313e-02, -6.8376e-02, -1.5578e-02,  8.3553e-02]],

    [[-3.2033e-02,  4.3956e-02, -2.1736e-02, -2.1183e-02],
     [ 4.7098e-02, -2.8876e-02,  6.4151e-02,  6.7120e-02],
     [ 7.9005e-03, -1.6685e-02, -6.9451e-02,  1.0795e-02],
     [ 1.4197e-02, -8.5314e-02,  7.3085e-02, -2.1615e-02],
     [-2.4433e-02,  1.6147e-02, -2.6234e-02, -6.5402e-02],
     [ 5.0357e-02, -2.3393e-03,  5.6049e-02,  3.7115e-02],
     [ 2.6990e-02, -3.2498e-02,  1.0601e-02, -1.1645e-02],
     [ 1.8832e-02, -5.5270e-02,  1.0874e-01, -2.3955e-02],
     [-6.4196e-03,  1.7824e-02, -7.6321e-03, -1.8961e-02],
     [-4.8489e-02,  9.4809e-03, -4.2953e-02, -1.8791e-02]],

    [[ 2.9793e-02, -1.1435e-02,  7.1202e-02,  3.8114e-02],
     [ 3.4568e-02,  5.5121e-02,  2.4039e-02, -1.0106e-01],
     [-5.7153e-02,  3.8913e-02,  1.8795e-02,  1.8338e-03],
     [-1.7938e-02,  1.3931e-02, -2.9045e-02, -1.6755e-02],
     [-1.0585e-02, -2.9855e-02,  5.7895e-03, -1.0541e-02],
     [ 7.5597e-03,  2.8414e-02,  1.1358e-04, -2.5612e-02],
     [ 5.6110e-02,  3.3737e-02,  5.0624e-03, -4.7735e-02],
     [ 4.3796e-02, -3.6556e-03, -7.6050e-02, -4.1810e-02],
```

```
          [ 5.3100e-03, -7.7142e-03,  5.0428e-02, -1.1749e-03],
          [-6.5851e-02, -3.5711e-02,  2.0275e-02,  1.1695e-02]],

         [[-9.8326e-02, -2.5781e-02,  2.3171e-02,  4.8900e-02],
          [ 5.2090e-02,  1.0216e-01, -9.3877e-03, -2.6939e-02],
          [-7.6315e-03,  2.5220e-02,  3.7234e-02, -7.6781e-02],
          [ 7.4216e-02, -3.7519e-02, -6.5198e-02,  1.4135e-02],
          [-4.4565e-02,  4.9714e-02,  5.3041e-02,  5.9862e-02],
          [ 2.3508e-02,  2.9700e-02, -1.2778e-02, -1.7595e-02],
          [-1.1837e-02,  5.0877e-02, -2.0870e-02, -1.2668e-02],
          [ 4.1511e-02, -3.0298e-02, -5.2912e-02,  2.5892e-02],
          [-5.1517e-02,  1.6017e-03,  4.1308e-02,  1.6990e-02],
          [-5.0386e-02, -4.0327e-02,  6.7870e-03, -7.2765e-04]],

         [[ 1.8464e-02, -3.0615e-02, -1.1375e-01, -1.0925e-02],
          [ 4.6758e-02,  9.2669e-02,  2.9211e-01, -8.4722e-03],
          [ 2.9926e-04,  5.8871e-02,  4.4926e-02,  9.9094e-03],
          [-3.3315e-03,  1.0131e-02,  8.2669e-02, -5.3462e-02],
          [-3.0162e-02, -4.8233e-02, -1.9366e-01,  5.8879e-03],
          [-1.3493e-02,  3.2501e-02,  1.2985e-01, -3.0696e-02],
          [ 4.6715e-02, -1.4954e-03,  1.7021e-01, -1.0984e-02],
          [ 4.0969e-02,  3.2787e-02,  6.8180e-02, -8.9982e-02],
          [ 2.9320e-02, -1.7175e-02, -7.7663e-02,  3.6064e-02],
          [-2.2414e-02, -5.4286e-02, -1.3325e-01,  5.6800e-02]],

         [[-1.0166e-02, -2.7691e-02,  4.5585e-02,  6.9049e-02],
          [ 3.5743e-02,  1.2174e-01,  1.2520e-02, -1.4579e-01],
          [ 3.3914e-02,  1.9404e-02, -3.9463e-02, -1.8352e-02],
          [ 1.7090e-02,  7.2636e-02,  4.3980e-02, -4.0188e-02],
          [-2.6315e-02, -5.4174e-02, -1.4394e-02,  5.3422e-02],
          [ 4.2551e-02,  5.6379e-02, -9.7179e-04, -6.3451e-02],
          [ 5.2552e-02,  3.9779e-02, -1.6789e-02, -2.7893e-02],
          [ 3.3845e-02,  8.8931e-02,  1.4214e-01, -2.6241e-02],
          [-8.7452e-02, -5.2818e-02, -4.3340e-02,  3.3481e-02],
          [-2.0295e-02, -9.0326e-02, -1.5096e-02,  4.6770e-02]]],
       device='cuda:0', requires_grad=True)
Parameter containing:
tensor([[[-0.0039, -0.0160, -0.0296,  0.1332]],

        [[ 0.0050,  0.0261, -0.0269, -0.0687]],

        [[ 0.0344,  0.0617,  0.0189, -0.1052]],

        [[ 0.0665,  0.0071, -0.0816, -0.0759]],

        [[-0.0142, -0.0032, -0.0010,  0.0670]],

        [[ 0.0356,  0.0161, -0.0227, -0.0972]],

        [[ 0.0081,  0.0044,  0.0180, -0.0618]],

        [[ 0.0092, -0.0211, -0.0476, -0.0899]],

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         [[ 0.0555,  0.0041, -0.0509,  0.0500]],

         [[-0.0260, -0.0234, -0.0181,  0.0514]]], device='cuda:0',
        requires_grad=True)
==========================================================================================
Layer (type:depth-idx)                             Output Shape              Param #
==========================================================================================
CPIKANModel                                        [1, 1]                    --
├─ModuleList: 1-1                                    --                        --
│    └─0.weight                                                                ├─40
│    └─1.weight                                                                ├─400
│    └─2.weight                                                                └─40
│    └─ChebyKANLayer: 2-1                              [1, 10]                   40
│    │    └─weight                                                              └─40
│    └─ChebyKANLayer: 2-2                              [1, 10]                   400
│    │    └─weight                                                              └─400
│    └─ChebyKANLayer: 2-3                              [1, 1]                    40
│    │    └─weight                                                              └─40
==========================================================================================
Total params: 480
Trainable params: 480
Non-trainable params: 0
Total mult-adds (Units.MEGABYTES): 0.00
==========================================================================================
Input size (MB): 0.00
Forward/backward pass size (MB): 0.00
Params size (MB): 0.00
Estimated Total Size (MB): 0.00
==========================================================================================

进程已结束，退出代码为 0

```